

BREST BRETAGNE AP, FranceWMO: **071100**Lat: **48.4441N**Lon: **4.4120W**Elev: **325**StdP: **14.52**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	29.5	32.0	23.5	17.9	32.8	26.5	20.7	35.1	32.0	49.4	28.2	49.1	5.9	80	0.588

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	78.5	66.1	74.7	64.3	71.5	62.9	67.7	75.7	65.8	71.6	64.3	68.8	9.0	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.9	93.3	69.7	63.7	89.4	67.8	62.5	85.6	66.1	32.2	76.0	30.7	71.8	29.6	68.8	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.8	22.9	20.4	DB	25.0	85.4	2.9	4.3	22.9	88.4	21.2	90.9	19.6	93.3	17.4	96.4	(4)
(5)			WB	23.9	71.1	2.8	2.7	21.9	73.0	20.3	74.6	18.7	76.1	16.7	78.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.9	44.7	44.7	47.1	49.6	54.6	59.2	62.3	62.3	59.3	55.1	49.4	45.7	(6)	
	DBStd	8.00	5.41	4.92	4.72	4.60	4.63	4.38	3.78	3.68	4.17	4.73	4.62	5.67	(7)	
	HDD50	742	174	155	112	60	11	0	0	0	0	11	65	154	(8)	
	HDD65	4478	630	569	554	461	325	182	101	102	178	308	469	599	(9)	
	CDD50	1790	8	6	23	49	152	277	382	380	279	169	46	19	(10)	
	CDD65	54	0	0	0	0	2	9	19	17	7	0	0	0	(11)	
	CDH74	389	0	0	0	1	20	92	141	93	39	3	0	0	(12)	
	CDH80	66	0	0	0	0	2	20	22	19	3	0	0	0	(13)	
(14) Wind	WSAvg	9.7	11.0	10.8	10.5	9.9	9.4	9.0	8.7	8.2	8.4	9.5	10.0	10.7	(14)	
(15) Precipitation	PrecAvg	45.7	5.5	4.4	3.7	3.2	3.0	2.3	2.3	2.7	3.1	4.6	5.3	5.6	(15)	
	PrecMax	63.2	11.3	10.5	8.7	8.7	6.3	6.3	6.0	6.4	6.7	10.0	11.0	13.0	(16)	
	PrecMin	30.9	0.7	0.4	0.6	0.2	0.4	0.1	0.4	0.4	0.4	0.4	1.3	0.9	(17)	
	PrecStd	7.1	2.6	2.2	2.0	2.0	1.5	1.5	1.4	1.6	1.7	2.2	2.4	2.7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	58.9	66.6	72.1	76.9	82.9	82.5	82.1	78.7	70.3	61.7	57.6	(19)	
		MCWB	53.2	51.4	54.8	57.9	63.0	68.1	68.5	68.3	66.3	62.7	57.9	54.9	(20)	
	2%	DB	54.4	54.3	59.7	65.0	70.4	75.8	77.9	75.6	73.6	66.2	59.2	56.2	(21)	
		MCWB	52.6	50.8	52.0	55.5	60.2	64.1	65.9	66.3	64.0	60.7	56.7	54.3	(22)	
	5%	DB	53.3	52.9	55.8	60.0	66.4	71.3	73.9	71.8	69.6	63.7	57.6	55.1	(23)	
		MCWB	51.7	50.5	51.0	52.9	58.5	62.0	64.1	64.4	62.1	59.8	55.5	53.5	(24)	
	10%	DB	52.1	51.8	53.6	56.9	62.7	67.5	70.1	69.0	66.5	61.9	56.5	53.9	(25)	
		MCWB	50.4	49.9	50.6	51.5	56.5	60.1	62.5	62.6	60.9	58.9	54.4	52.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.5	53.7	55.8	59.8	64.5	69.4	70.0	70.6	68.2	63.9	59.5	56.1	(27)	
		MCDB	55.0	55.6	63.3	67.9	72.6	80.3	79.7	78.3	76.1	67.8	60.3	56.6	(28)	
	2%	WB	53.3	52.3	53.7	56.1	61.3	65.6	67.2	67.5	65.4	62.2	57.5	54.9	(29)	
		MCDB	54.1	53.5	57.1	62.9	68.5	73.2	75.8	73.4	70.3	64.6	58.6	55.7	(30)	
	5%	WB	52.2	51.3	52.5	54.0	59.3	63.0	65.1	65.6	63.6	60.8	56.1	53.9	(31)	
		MCDB	53.0	52.5	54.9	58.3	65.3	69.4	71.2	69.9	67.3	63.0	57.4	54.9	(32)	
	10%	WB	50.9	50.2	51.3	52.6	57.3	61.0	63.4	64.1	62.2	59.6	54.7	52.7	(33)	
		MCDB	51.9	51.6	53.2	56.0	61.4	65.8	68.1	67.4	65.3	61.5	56.2	53.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	10.0	11.5	13.2	13.4	13.6	13.5	13.6	14.0	11.8	9.7	9.3	(35)	
		MCDBR	7.9	10.7	15.4	19.5	19.9	21.5	21.2	18.4	18.9	13.3	9.9	7.6	(36)	
	5% WB	MCWBR	7.8	8.7	9.9	11.4	11.7	11.4	10.9	10.0	10.5	8.7	8.2	7.4	(37)	
		MCWBR	7.3	8.8	12.2	15.6	17.8	18.5	17.8	15.2	15.4	10.9	8.9	7.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.334	0.348	0.373	0.381	0.385	0.378	0.379	0.368	0.358	0.355	0.335	0.327	(40)	
		taud	2.396	2.379	2.324	2.321	2.355	2.399	2.400	2.451	2.462	2.456	2.425	2.364	(41)	
	Ebn at Noon		229	254	265	275	278	280	278	277	269	249	227	213	(42)	
		Edh at Noon	22	27	34	37	37	36	36	32	29	25	21	20	(43)	
(44) All-Sky Solar Radiation	RadAvg		305	561	935	1466	1738	1881	1787	1531	1221	718	382	254	(44)	
	RadStd		23	58	96	122	152	197	97	125	87	53	30	30	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (23 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page